Solar Farm Project Finance Model

Investment summary report for a simplified 50 MW renewable infrastructure asset

Metric	Base case
Total capex	£37.5m
Debt / equity	65% / 35%
Minimum DSCR	1.09x
LLCR	1.37x
Project IRR	7.8%
Equity IRR	10.0%

Prepared as a portfolio project to demonstrate infrastructure finance modelling, renewable energy cash flow forecasting and lender-focused debt analysis.

1. Executive Summary

This report presents a simplified project finance model for a 50 MW solar farm. The model forecasts construction funding, electricity generation, operating cash flows, tax, debt service and investor returns across a 25-year project life. It is designed as a practical portfolio project rather than a live investment recommendation.

The base case produces an equity IRR of **10.0%**, a project IRR of **7.8%**, a minimum DSCR of **1.09x** and an LLCR of **1.37x**. The project is financeable under the base case, but it is sensitive to lower generation, lower achieved power price, higher capex and higher interest rates.

Output	Base case result	Interpretation
Total capex	£37.5m	Construction funding required
Total debt	£24.4m	Senior debt funded at 65% gearing
Total equity	£13.1m	Sponsor equity funded at 35% gearing
Minimum DSCR	1.09x	Lowest annual debt coverage during repayment period
Average DSCR	1.55x	Average annual debt coverage during repayment period
LLCR	1.37x	PV of loan-life CFADS divided by outstanding debt
Project IRR	7.8%	Unlevered project return
Equity IRR	10.0%	Levered return to equity investors

2. Project Context

Solar PV is a major source of renewable capacity growth. The IEA expects solar PV and wind to account for most renewable capacity additions through 2030, supported by lower generation costs and policy support in many markets. This makes solar a useful asset type for learning infrastructure finance, because it combines a real physical asset, long-term operating cash flows and lender-driven risk analysis.

In project finance, lenders focus heavily on the cash flows generated by the project asset itself. World Bank PPP guidance explains that project finance lenders rely on the project company and project cash flows for debt repayment. This is why the model centres on CFADS, DSCR and LLCR rather than only accounting profit.

3. Model Objective

The model answers four practical finance questions:

Question	Model output used
How much funding does the project need?	Total capex, debt drawdown and equity contribution
Can the project repay lenders?	CFADS, debt service, DSCR and LLCR
What return does equity earn?	Equity cash flows and equity IRR
What happens under stress?	Downside, base and upside scenario cases

4. Base Case Assumptions

Category	Assumption	Value
Asset	Capacity	50 MW
Asset	Project life	25 years
Construction	Construction period	1 year
Construction	Capex per MW	£750,000
Operations	Capacity factor	17.0%
Operations	Annual degradation	0.5%
Revenue	PPA-style starting tariff	£65/MWh

Category	Assumption	Value
Revenue	PPA escalation	2.0% per year
Costs	Opex per MW	£15,000/MW/year
Costs	Opex escalation	2.0% per year
Financing	Debt / equity	65% / 35%
Financing	Interest rate	6.5%
Financing	Debt tenor	15 years
Tax	Tax rate	25%

5. Model Structure

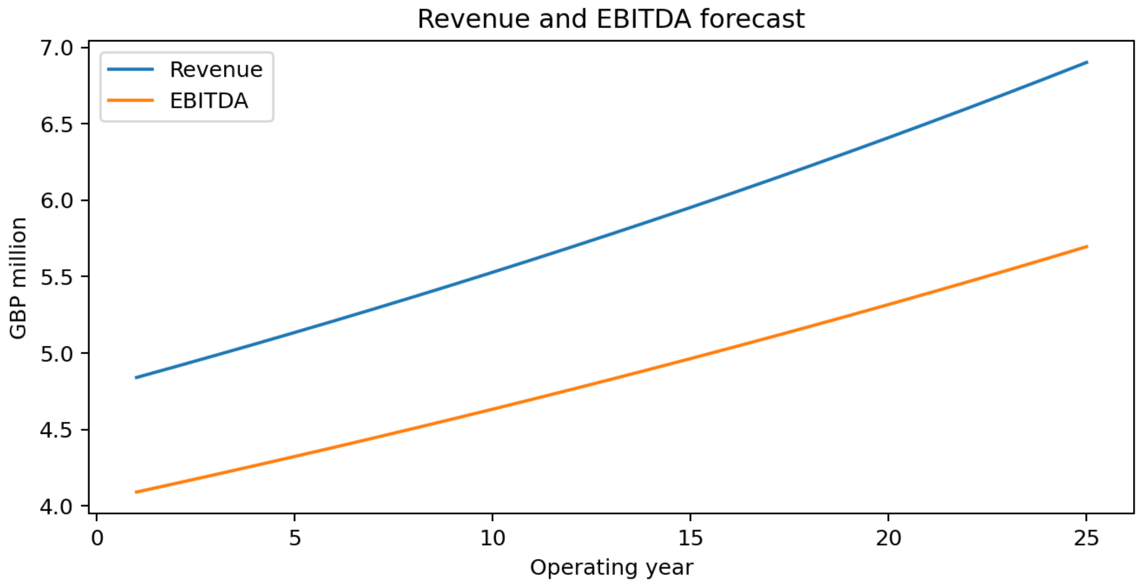
The workbook is structured to keep inputs, calculations and outputs separate. This makes the model easier to audit and easier to explain in an interview.

Tab	Purpose
00 Cover	Front page with project summary and headline outputs
01 Inputs	Main assumptions for project life, capex, operations, revenue and financing
02 Timeline	Year sequence and flags for construction, operations and debt repayment
03 Construction	Capex spend, debt drawdown and equity contribution
04 Operations	Generation, PPA price, revenue, opex and EBITDA
05 Debt	Opening debt, interest, principal repayment and closing debt
06 Tax	Simplified taxable income and tax charge
07 Cash Flow	CFADS, debt service, cash after debt and equity cash flow
08 Returns	DSCR, LLCR, project IRR and equity IRR
09 Sensitivities	Downside, base and upside scenario comparison
10 Dashboard	Charts and summary outputs
11 Checks	Model integrity checks

6. Operating Forecast

Revenue is calculated as net generation multiplied by the PPA-style tariff. Generation starts in Year 1, declines with annual degradation and is then multiplied by the escalating power price. EBITDA is revenue minus operating costs.

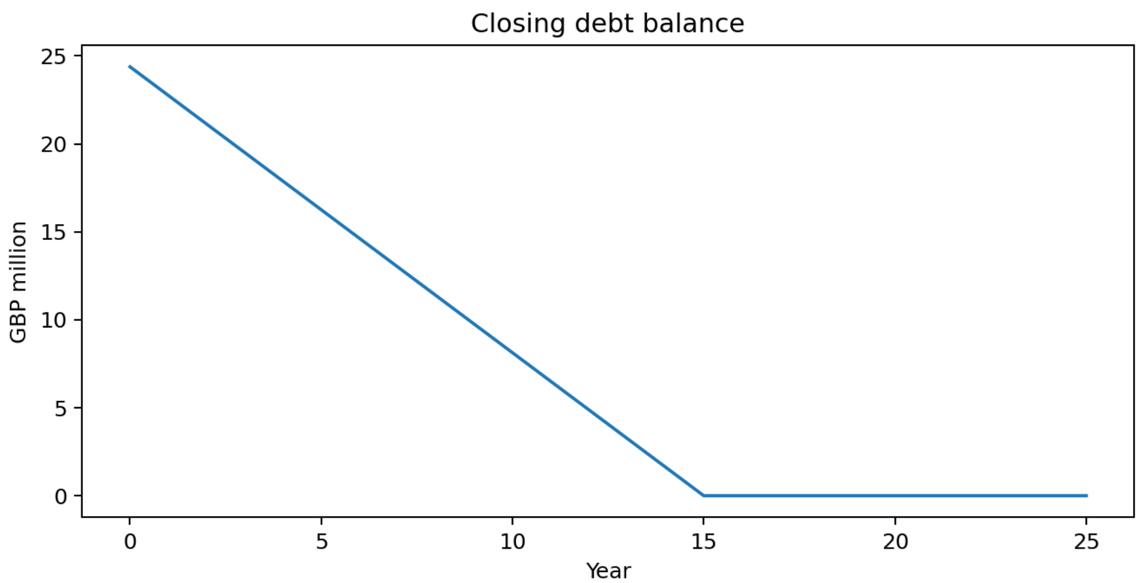
Year	Generation (MWh)	PPA price (£/MWh)	Revenue (£m)	Opex (£m)	EBITDA (£m)
1	74,460	65.00	4.84	0.75	4.09
2	74,088	66.30	4.91	0.77	4.15
3	73,717	67.63	4.99	0.78	4.20
4	73,349	68.98	5.06	0.80	4.26
5	72,982	70.36	5.13	0.81	4.32
10	71,176	77.68	5.53	0.90	4.63
15	69,414	85.77	5.95	0.99	4.96
20	67,696	94.69	6.41	1.09	5.32
25	66,020	104.55	6.90	1.21	5.70

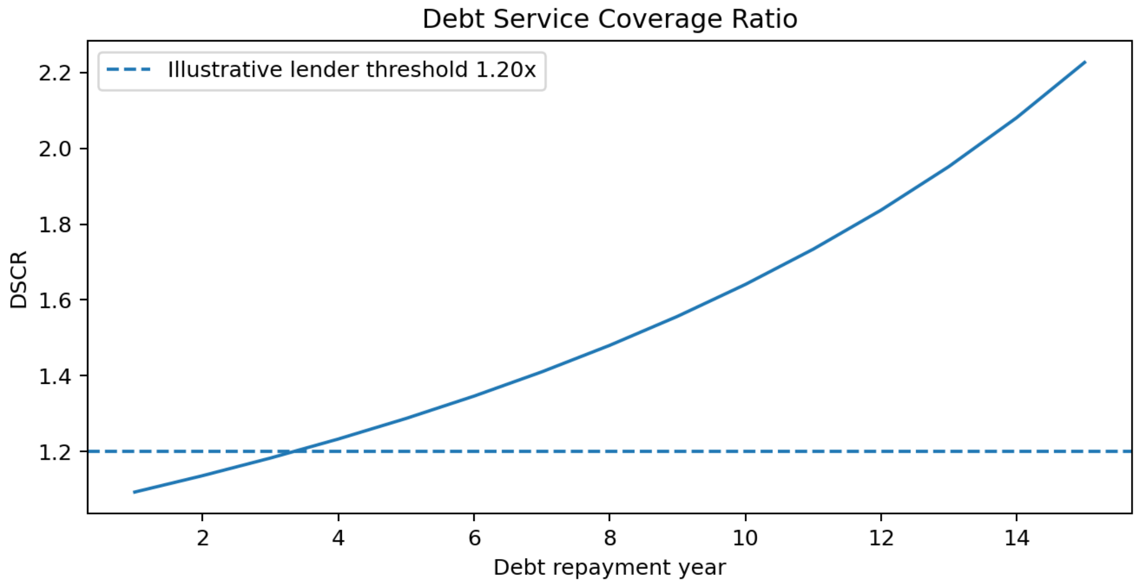


7. Debt and Coverage Analysis

The model uses a simple straight-line principal repayment profile across a 15-year debt tenor. DSCR measures the annual ability of CFADS to pay interest and principal. LLCR compares the present value of future loan-life CFADS against outstanding debt.

Year	CFADS (£m)	Debt service (£m)	DSCR	Closing debt (£m)
1	3.45	3.16	1.09x	22.75
2	3.47	3.05	1.14x	21.12
3	3.48	2.95	1.18x	19.50
4	3.50	2.84	1.23x	17.88
5	3.52	2.73	1.29x	16.25
10	3.62	2.21	1.64x	8.12
15	3.74	1.68	2.23x	0.00





8. Scenario Analysis

Scenario analysis tests how the model performs when key assumptions change. This is important because renewable infrastructure returns can be affected by construction cost, energy yield, tariff level and financing cost.

Scenario	Capex	PPA price	Generation	Interest rate	Min DSCR	LLCR	Equity IRR
Downside	+10%	-10%	-5%	+1.0%	0.81x	1.00x	4.7%
Base	0%	0%	0%	6.5%	1.09x	1.37x	10.0%
Upside	-5%	+10%	+5%	-0.5%	1.37x	1.74x	15.0%

The downside case falls below a comfortable lender position, with minimum DSCR below 1.00x and LLCR close to 1.00x. This shows why lenders stress test renewable projects before agreeing debt sizing. The upside case benefits from stronger generation and pricing as well as lower capex and financing cost.

9. Key Risks

Risk	Model impact	Mitigation / comment
Generation risk	Lower energy output reduces revenue and CFADS	Use conservative P50/P90 yield cases and technical due diligence
PPA price risk	Lower realised price reduces debt capacity and equity returns	Use contracted PPA terms or price sensitivities
Capex overrun	Higher construction cost increases funding need and weakens returns	Use contingency, fixed-price EPC contract and sponsor support
Interest rate risk	Higher interest increases debt service and reduces DSCR	Use hedging or conservative debt sizing
Operating cost risk	Higher opex reduces EBITDA and CFADS	Use O&M contract assumptions and inflation escalation
Model simplification risk	Outputs may overstate precision	Clearly disclose simplified tax, debt and reserve assumptions

10. Limitations

This model is intentionally simplified for learning and portfolio use. It does not include depreciation schedules, tax loss carry-forwards, debt service reserve accounts, maintenance reserves, debt sculpting, refinancing, working capital, connection costs, curtailment, merchant tail pricing or detailed legal documentation. Those items would be needed for a professional-grade transaction model.

The model also uses conventional project finance debt for learning purposes. A future extension could compare the base case with a sukuk-style asset-backed structure to make the project more aligned with Islamic finance principles.

11. Portfolio Value

This project shows that the candidate can connect engineering-style asset assumptions with finance outputs. It is directly relevant to infrastructure finance, project finance, renewable energy investment, energy transition roles and technical finance positions that require structured cash flow analysis.

CV bullet: Built a solar project finance model in Excel to forecast construction costs, operating cash flows, CFADS, debt repayment, DSCR, LLCR and equity IRR under base, downside and upside scenarios.

12. References

Source	Use in report
World Bank PPP Resource Center	Project Finance - Key Concepts. Used for project finance structure and lender reliance on project cash flows.
World Bank PPP Resource Center	Key Issues in Developing Project Financed Transactions. Used for DSCR and LLCR context.
IEA	Renewables 2024. Used for renewable capacity and solar PV market context.
NREL	Utility-Scale PV: 2023 Annual Technology Baseline. Used for utility-scale PV fixed O&M benchmark context.
IRENA	Renewable Power Generation Costs in 2023. Used for high-level solar PV cost competitiveness context.

Disclaimer: This report is for educational and portfolio purposes only. It is not investment, legal, tax or Sharia advice.